

TITLE PAGE

Problem Statement ID –

Q1

Problem Statement Title –

Eco-Loop Building Agents

Theme –

AI / Sustainability / Smart Automation

PS Category –

Software

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GitHub Repository:

<https://github.com/Sufi-S/Eco-Loop-Building-Agent>

Demo Video:

https://drive.google.com/file/d/1FWK5dN9UXB_YNnTK2QWa78lhUjQ_eZWJ/view?usp=sharing

IDEA TITLE — Eco-Loop Building Agents

Proposed Solution

An autonomous AI agent that continuously reads a building's live simulated state (zone temperatures, humidity, occupancy, energy use) and adjusts HVAC setpoints to minimize energy consumption while preserving occupant comfort — a genuine closed feedback loop, not a one-time optimization.

Detailed Explanation

- EnergyPlus simulates a real 6-zone office building (DOE reference model, Chicago climate)
- A Python controller parses the simulation output into structured data
- An MCP-style tool layer exposes building state and actions as callable functions
- An LLM agent (AWS Bedrock, Claude Sonnet 4.6) reasons over this state and decides setpoint changes, with natural-language justification
- Decisions are written back into the building model and re-simulated to verify real energy impact

How it Addresses the Problem / Innovation

Unlike static rule-based building management systems, the agent reasons in natural language about real physical conditions and explains its decisions — making it explainable, auditable, and adaptable, while a real physics simulator (not a toy model) verifies every decision before it's trusted.

TECHNICAL APPROACH

Technologies Used

- EnergyPlus — building energy simulation engine
- Python — orchestration, parsing (pandas), simulation control
- AWS Bedrock (Claude Sonnet 4.6) — LLM-based decision-making agent
- Streamlit + Plotly — results dashboard

Methodology — The Closed Loop



Loops back to EnergyPlus continuously — a genuine closed feedback loop

Fallback design: agent runs in rule-based mock mode if Bedrock is unavailable, ensuring the loop always completes.

FEASIBILITY AND VIABILITY

Feasibility

Fully built and demonstrated end-to-end using free/low-cost tools (EnergyPlus is open-source, AWS Bedrock is pay-per-call). Achieved 5.8% verified energy savings on a real DOE reference building model, with comfort preserved — proving the approach works on realistic physics, not a toy simulation.

Potential Challenges & Risks

- Shared HVAC schedules across zones limit per-zone independence in some buildings
- LLM API costs and latency at scale (many buildings, frequent decisions)
- Requires real occupancy/sensor data integration for live deployment (vs. simulated schedules)

Strategies for Overcoming Challenges

- Rewrite building models to use per-zone schedules for finer control
- Batch decisions at longer intervals (hourly, not per-second) to control LLM costs
- Integrate with existing BMS/IoT sensor platforms for live occupancy data

ARTIFACTS

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Real AI agent reasoning (AWS Bedrock, Claude Sonnet 4.6):

"This occupied perimeter zone is at 26.9°C, exceeding comfortable temperature thresholds. Cooling is needed to maintain occupant comfort."

Results Summary:

Baseline

518,798.27 kWh

Optimized

488,552.72 kWh

Savings

5.8% (30,245.55 kWh)

Eco-Loop Building Agents — AI-Optimized Energy Dashboard

A closed-loop system that uses EnergyPlus simulation, a Python controller, and an AI agent (Bedrock LLM) to adjust setpoints and reduce energy consumption while maintaining occupant comfort.

Baseline Total Energy

518,798.27 kWh

Optimized Total Energy

488,552.72 kWh

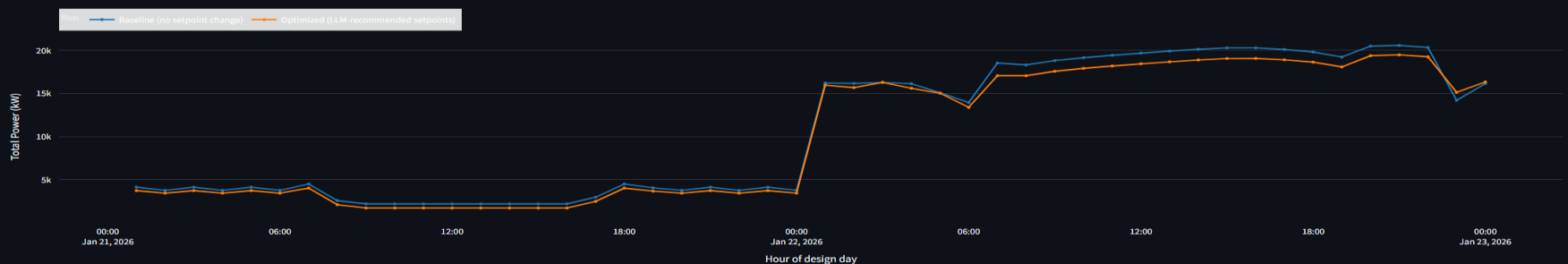
↓ 30,245.55 kWh

Energy Savings

30,245.55 kWh

↑ 5.8%

Total Electricity Demand Over Time



RESEARCH AND REFERENCES

- EnergyPlus documentation: <https://energyplus.net>
- DOE Commercial Reference Buildings: RefBldgSmallOfficeNew2004
- AWS Bedrock documentation: <https://docs.aws.amazon.com/bedrock/>
- Anthropic Claude models: <https://www.anthropic.com/claude>
- ASHRAE Standard 55 — Thermal Environmental Conditions for Human Occupancy
- Model Context Protocol (MCP) pattern reference: <https://modelcontextprotocol.io>
- Project repository: <https://github.com/Sufi-S/Eco-Loop-Building-Agent>